Computer Vision Practical Report:

SHARPEN IMAGE USING A LAPLACIAN FILTER WITH NEGATIVE CENTER COEFFICIENT

1. AIM

To sharpen an image using a Laplacian filter with a negative center coefficient in OpenCV, display the original and sharpened images, and save the result to disk.

2. PROCEDURE

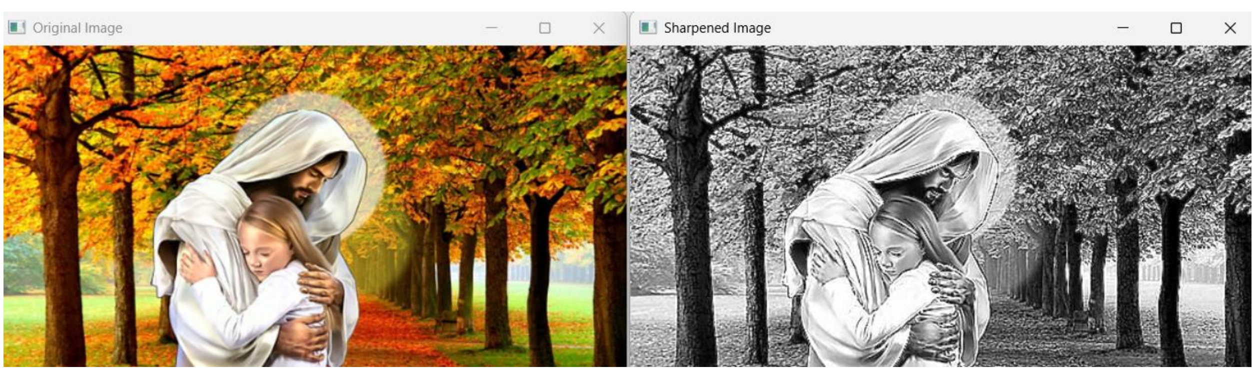
- Load the image and convert it to grayscale.
- Define a 3×3 Laplacian kernel with a negative center value (e.g., center = -4 or -8, edges = $+1$).
- Filter the image using cv2.filter2D() to obtain the Laplacian edge map.
- Subtract the Laplacian response from the original image to create the sharpened output.
- Display the original and sharpened images side-by-side using Matplotlib, and save the result using cv2.imwrite().

3. PROGRAM CODE

```
import cv2
import numpy as np

image = cv2.imread(r"C:\Users\adminuser\Desktop\Python CV\E20-IMG.png")

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Laplacian mask with negative center coefficient
kernel = np.array([
    [0, 1, 0],
    [1, -4, 1],
    [0, 1, 0]
```

```
], dtype=np.float32)

# Apply Laplacian mask
laplacian = cv2.filter2D(gray, cv2.CV_32F, kernel)

# Sharpen the image
sharpened = gray - laplacian

# Convert to 8-bit
sharpened = cv2.convertScaleAbs(sharpened)

cv2.imshow("Original Image", image)
cv2.imshow("Sharpened Image", sharpened)

cv2.imwrite("laplacian_sharpened.jpg", sharpened)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.